Macroscopic Dephasing in LMT Atom Interferometry from an Operational Coherence-Cell Bridge

A Short Letter on the NDK Momentum-Quadratic Contrast Floor

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Abstract

A central open question in macroscopic quantum mechanics is whether contrast loss in large-momentum-transfer (LMT) atom interferometry remains entirely reducible to environmental and technical noise, or whether a genuinely irreducible geometric floor emerges once momentum separation and interrogation time become sufficiently large. In the Null-Directional Kinematics (NDK) framework, this question is sharpened by a finite-throughput transport picture of the directional vacuum. The present Letter develops a compact phenomenological bridge between near-term atom interferometers and the cosmological strain scale of that framework, calibrated in the broader NDK program against the high-redshift supermassive-black-hole collapse envelope of [1]. The bridge combines a cell-level transport burden inherited from the reduced NDK Fisher transport sector with an operational pre-split coherence-cell prescription for laboratory transfer. Recent action-scale closure work sharpens the upstream normalization by isolating the primitive cell as the unique carrier of the absolute action scale, so that the Planck-anchored transport coefficient is no longer introduced as a free ultraviolet dimensional guess. What remains phenomenological is the coarse-graining from that closure-informed substrate normalization to the operational source-level coherence cell probed by an atom interferometer.

The resulting leading-order prediction is a residual dephasing law

$$D_{\mathrm{pred}} \equiv -\ln V_{\mathrm{res}} \propto T\, n^2 k_{\mathrm{eff}}^2,$$

with normalization controlled by the ratio of the interferometric coherence-support width to the pre-split coherence length of the atomic source. A benchmark together with a broader bridge analysis is used to compare current $n \sim 10^2$ platforms to the $n \sim 10^3$ regime relevant to MAGIS-like architectures, while the effective threshold ratio is held fixed at the representative cross-program benchmark value $\mathcal{I}_*/A \approx 3.28$, adopted here as a phenomenological input rather than as a laboratory fit parameter. Within the operational coherence-cell bridge developed here, current $n \sim 10^2$ platforms remain parametrically lower, while long-time thermal operating points in the $n \sim 10^3$ regime move into the sub-percent sensitivity window and, for broader representative coherence-support widths, can approach the 10^{-3} contrast-loss benchmark. The experimental discriminator is therefore sharp: at fixed interrogation time, the bridge predicts a residual contrast-loss component with leading quadratic dependence on LMT order, and, under a thermal identification of the pre-split coherence cell, a quadratic dephasing coefficient that scales linearly with source temperature. The focused figure shown here should be read as a model-side width-sensitivity band at a chosen long-time thermal operating point rather than as a statistical experimental uncertainty interval or a full multi-parameter sweep.

1 Introduction

The empirical success of quantum mechanics rests on the linear superposition principle, yet the question of whether macroscopic coherence is fundamentally unlimited remains unresolved. In practice, visibility loss in matter-wave interferometry is modeled through environmental coupling, finite pulse fidelity, laser phase noise, wavefront effects, background gas scattering, expansion of the cloud, and other apparatus-specific systematics. At the same time, the continued development of large-momentum-transfer (LMT) atom interferometry pushes experiments into regimes where path separation, coherence time, and phase-space burden grow dramatically [2–4].

This makes long-baseline instruments especially interesting as discriminators of any residual dephasing law that is intrinsic rather than merely technical. MAGIS-100, for example, is designed as a 100 m-scale atom interferometer and as a pathfinder for precisely the large-T, large-LMT control problem relevant to new-physics searches and future long-baseline interferometers [4,5]. The present Letter asks whether such architectures should be expected to encounter a residual contrast floor whose leading dependence is not merely linear in pulse number or interrogation time, but instead scales with the geometric combination $T(\Delta p)^2$.

In the broader NDK framework, the directional vacuum is modeled as a finite-throughput informational substrate, and its transport sector is governed by Fisher information in the privileged amplitude representation $P = |\Psi|^2$ [6, 7]. Recent work has derived the reduced Schrödinger transport sector of that framework in a controlled low-energy limit, showing that the matter-wave phase sector arises as the reversible transport completion of the reduced Fisher-density theory rather than as an independent postulate [8]. More recently, the action-scale closure problem has been localized to the primitive-cell sector, which lies outside the reduced one-particle scaling symmetry and uniquely carries the absolute action normalization [9]. The specific goal of the present Letter is therefore narrower and more practical: to formulate a compact phenomenological bridge between that now-derived, closure-informed transport law and LMT atom interferometry, and then to show what that bridge implies for current $n \sim 10^2$ instruments versus future $n \sim 10^3$ regimes.

The main result is a closure-informed operational coherence-cell bridge in which the transport burden grows kinematically with $Tn^2k_{\rm eff}^2$, while the experimentally active source-level coherence cell is tied at leading order to the pre-split source coherence rather than to the post-splitting momentum burden itself. This separation preserves the momentum-quadratic signature and yields a direct experimental forecast. In broad benchmark form, the bridge permits substantial upward movement of the residual in the $n \sim 10^3$ regime for favorable coherence-transfer assumptions, while the focused thermal illustration presented here adopts a narrower long-time operating point and shows a representative width-sensitivity band that remains substantially lower, with current $n \sim 10^2$ platforms still parametrically below the relevant sensitivity window.

2 The phenomenological bridge

2.1 Transport burden

The first ingredient is a dimensionless transport-burden variable,

$$\mathcal{D}_{\rm tr} \equiv \pi\,\chi\,\mathcal{I}_*/A\,c\,\ell_{\rm P}\,T\,n^2k_{\rm eff}^2, \tag{1}$$

where T is the interrogation time, n is the effective LMT order, $k_{\rm eff}$ is the effective wavevector, $\ell_{\rm P}$ is the Planck length, and χ is an order-unity geometric factor. The factor $\mathcal{I}_*/A$ is the cosmologically calibrated directional-strain threshold inherited from the broader NDK program [6], and is fixed

phenomenologically in the SMBH-envelope analysis of [1] by the early-universe supermassive-black-hole collapse envelope, anchored to high-redshift quasar data such as [10], rather than by laboratory data. In the present Letter, this factor is not treated as a swept laboratory parameter, but is fixed to the representative benchmark value $\mathcal{I}_*/A \approx 3.28$ inferred from that cosmological calibration. This benchmark is used here as an adopted phenomenological input rather than as a claimed exact first-principles universal constant. In that sense, Eq. (1) should be interpreted as the accumulated Fisher-transport burden carried by a split matter-wave configuration, normalized to the same effective geometric-capacity scale used in the current cosmological realization of the broader NDK program. A successful laboratory test of the coherence-cell bridge would therefore support the transport scaling developed here more directly than it would, by itself, validate the full phenomenological construction of the SMBH-envelope calibration from which the benchmark value is imported.

The geometric factor χ is retained as an order-unity post-reduction parameter. A naive isotropic projection from the S^2 bundle to the beam axis would yield a bare overlap of 1/3, but the angular reduction leading to the emergent transport law already partially absorbs this sphere-to-axis averaging. The residual χ entering the phenomenological bridge is therefore not simply the bare projection factor, and the benchmark $\chi = 1$ is adopted here as a post-reduction effective coupling.

Recent action-scale closure results do not by themselves replace the threshold ratio $\mathcal{I}_*/A$ used here [9]. They fix the primitive action normalization of the substrate, whereas $\mathcal{I}_*/A$ in the present Letter continues to encode the separate threshold burden imported from the broader NDK strain calibration. The present update therefore sharpens the status of the Planck-anchored transport coefficient without eliminating the phenomenological benchmark $\mathcal{I}_*/A \approx 3.28$ used in the forecast.

The importance of Eq. (1) is structural: it isolates a kinematic quantity that grows linearly with T and quadratically with momentum separation, but does not by itself yet specify the scale of the coherence-support cell on which that burden is registered.

2.2 Reduced derivation from the master action

A recent controlled low-energy reduction of the NDK master action has already closed the upstream gap from the sphere-bundle Fisher transport law to an emergent reduced Schrödinger sector [8]. What is not yet available is a fully microscopic reduction from that transport law to a concrete laboratory atom interferometer. The middle of the bridge can nevertheless be made more explicit in a weak-anisotropy coarse-grained limit.

Let the directional amplitude be expanded around the isotropic vacuum as

$$\psi(\hat{k}; x) = \frac{1}{\sqrt{4\pi}} \left[1 + u_i(x)\hat{k}^i + q_{ij}(x)\Big(\hat{k}^i\hat{k}^j - \tfrac{1}{3}\delta^{ij}\Big) + \cdots \right], \tag{2}$$

where u_i is the dipole moment and q_{ij} is the traceless quadrupole of the directional distribution. After integrating out the angular dependence, the lowest directional moments define an effective spacetime transport sector whose leading couplings to a nonrelativistic matter envelope $\Phi = \sqrt{\rho}\, e^{i\phi}$ enter through the matter current and stress. When the directional modes are integrated out in an adiabatic or coarse-grained approximation, the lowest positive-semidefinite local correction induced in the matter sector is a transport penalty quadratic in the phase gradient, of the schematic form

$$\Delta\mathcal{L}_{\text{eff}} = \zeta\, \mathcal{I}_*/A\, c\, \ell_{\text{P}}\, \rho\, |\nabla\phi|^2, \tag{3}$$

with ζ an order-unity reduction coefficient.

For a symmetric two-branch LMT configuration with relative wavevector separation $|\Delta\mathbf{k}| = nk_{\text{eff}}$, the excess phase-gradient burden relative to the unsplit state scales as $\chi\,(nk_{\text{eff}})^2$, where

χ absorbs branch weighting, projection geometry, and pulse-sequence details. Accumulating this burden over interrogation time T gives the cell-level transport variable

$$D_{\text{cell}} = \pi\,\chi\,\mathcal{I}_*/A\,c\,\ell_{\text{P}}\,T\,n^2 k_{\text{eff}}^2, \tag{4}$$

which is precisely the transport-burden variable $\mathcal{D}_{\text{tr}}$ defined in Eq. (1). In this sense, the form of $\mathcal{D}_{\text{tr}}$ is not introduced as a purely freestanding phenomenological postulate, but is instead the natural coarse-grained cell-level motivation suggested by the Fisher transport sector acting on a split matter-wave configuration in the weak-anisotropy limit.

2.3 Primitive-cell normalization and the remaining open bridge

The reduced derivation above explains why the cell-level burden is quadratic in the relative phase gradient, but it does not by itself determine the absolute action normalization of the reduced phase sector. Recent action-scale closure work sharpens that point structurally [9]. In that analysis, the primitive cell A_* is isolated as the unique substrate object lying outside the reduced one-particle scaling symmetry. One primitive phase cycle carries the normalized action

$$S_{\text{cell}} = \frac{c^3}{G} A_*, \tag{5}$$

and, under the phase-inheritance theorem of the reduction, the emergent reduced action scale satisfies

$$\hbar_{\text{eff}} = S_{\text{cell}}. \tag{6}$$

If one further adopts the gravitational-entropy consistency condition used in [9], then $A_* = \ell_P^2$, and the reduced action scale correspondingly recovers the observed quantum of action. In that sense, the Planck-anchored scale entering the present transport burden is no longer introduced as an arbitrary ultraviolet dimensional insertion, but as the closure-informed normalization inherited from the primitive-cell sector.

What remains open is a different question: how that fixed substrate normalization is coarse-grained into the operational pre-split coherence structure of a concrete atom source. The present Letter addresses precisely that transfer problem. The primitive action scale is treated as closure-informed, while the source-level active cell count remains a phenomenological bridge to experiment. This sharpened separation of roles is what motivates the operational coherence-cell bridge below.

2.4 Operational coherence-cell bridge

To avoid notational conflict with the primitive cell A_* appearing in the action-scale closure analysis, denote the *operational* pre-split coherence-cell area of the atomic source by

$$A_{\text{coh}} = \eta\,\lambda_{\text{coh},0}^2, \tag{7}$$

where $\lambda_{\text{coh},0}$ is the pre-split coherence length of the atomic source and η is an order-unity geometric factor.

In the absence of a fully microscopic derivation of the operational source-level cell scale, the natural leading phenomenological closure is to tie that scale to the pre-split coherence properties of the atomic source rather than to the later momentum burden imposed by the interferometer sequence. Since those coherence properties are determined at leading order by the thermodynamic preparation of the ensemble, the working identification $A_{\text{coh}} = \eta\,\lambda_{\text{coh},0}^2$ states that the effective

coherence cell is set before the LMT sequence begins, while the subsequent kinematic splitting tests whether that source-level cell can support a transport burden growing as $Tn^2k_{\rm eff}^2$.

This operational source-level cell should not be identified with the primitive substrate cell of [9]. The latter fixes the absolute action normalization upstream; the former is the coarse-grained laboratory cell through which that substrate normalization is interrogated experimentally. Accordingly, the remaining uncertainty in the present Letter should be read as a laboratory transfer uncertainty, not as an unfixed substrate action-scale ambiguity.

This separation of roles is the central laboratory transfer hypothesis of the Letter:

> *Thermodynamics fixes the pre-split coherence cell; kinematics stresses that cell through the accumulated transport burden.*

In the coherence-cell bridge adopted here, the residual dephasing is taken to be the transport burden registered on the ratio of the transverse coherence-support width squared to the operational pre-split coherence-cell area:

$$D_{\rm pred} \equiv \mathcal{D}_{\rm tr}\,\frac{w_{\rm coh}^2}{A_{\rm coh}}, \tag{8}$$

where $w_{\rm coh}$ is the transverse coherence-support width relevant to the evolving cloud. Using $A_{\rm coh} = \eta\,\lambda_{\rm coh,0}^2$, this gives

$$D_{\rm pred} \equiv -\ln V_{\rm res} = \frac{\pi\chi}{\eta}\,\mathcal{I}_*/A\,c\,\ell_{\rm P}\,T\,n^2k_{\rm eff}^2\left(\frac{w_{\rm coh}}{\lambda_{\rm coh,0}}\right)^2. \tag{9}$$

Equation (9) is the master phenomenological equation of the present Letter.

Several features are immediate.

First, the leading dependence is

$$D_{\rm pred} \propto T\,n^2k_{\rm eff}^2. \tag{10}$$

Thus, at fixed T, the residual dephasing scales quadratically with LMT order.

Second, the amplitude of the effect is controlled by the ratio $(w_{\rm coh}/\lambda_{\rm coh,0})^2$. This makes physical sense: an extended coherence-support region carried by a comparatively small pre-split coherence cell produces a larger effective burden.

Third, the bridge is explicitly phenomenological at the laboratory transfer step. The present Letter does not claim that Eq. (7) has already been derived microscopically from the full NDK transport kernel. The claim is more modest and more testable: if the pre-split source coherence sets the operational cell scale, then Eq. (9) is the natural lowest-order forward prediction.

2.5 Baseline benchmark

A representative benchmark, chosen to keep the geometry convention consistent with the focused thermal illustration shown later in figure 1, is obtained by taking

$$\mathcal{I}_*/A \approx 3.28, \qquad \chi = 1, \qquad \eta = \pi, \qquad k_{\rm eff} = 10^7\,{\rm m}^{-1}, \tag{11}$$

together with

$$w_{\rm coh} = 100\ \mu{\rm m}, \qquad \lambda_{\rm coh,0} = 3\ \mu{\rm m}. \tag{12}$$

Here $\mathcal{I}_*/A \approx 3.28$ is the adopted representative benchmark value used throughout the present Letter.

For a current-tech benchmark $T = 1.4$ s and $n = 100$, Eq. (9) gives

$$D_{\rm pred} \approx 2.47\times10^{-5}, \qquad V_{\rm res} = e^{-D_{\rm pred}} \approx 0.999975.$$

Thus the predicted residual dephasing remains far below the sub-percent level in this regime. By contrast, for a MAGIS-like benchmark $T = 3.0$ s and $n = 1000$, the same equation gives

$$D_{\rm pred} \approx 5.28 \times 10^{-3}, \qquad V_{\rm res} = e^{-D_{\rm pred}} \approx 0.9947.$$

This moves the prediction into the several-10^{-3} contrast-loss regime, above the 10^{-3} benchmark while still below the percent level. In this baseline realization, the theory therefore predicts a clear parametric separation between $n \sim 10^2$ and $n \sim 10^3$ architectures.

3 Parameter sweep

The broad bridge dependence is encoded in Eq. (9) and in the benchmark discussion above, but the focused illustration shown in figure 1 is intentionally narrower. Rather than displaying the full multi-parameter bridge sweep, the figure fixes a representative long-time thermal operating point and shows how the forward prediction varies under a model-side scan of the transverse coherence-support width alone.

Specifically, figure 1 adopts

$$T = 9\,\mathrm{s}, \qquad \Theta = 400\,\mathrm{pK}, \qquad k_{\rm eff} = 10^7\,\mathrm{m}^{-1}, \qquad \chi = 1, \qquad \eta = \pi, \qquad \mathcal{I}_*/A \approx 3.28, \qquad \xi_{\rm th} = 1,$$

and varies

$$w_{\rm coh} \in [30, 80]\,\mu\mathrm{m}$$

while holding the remaining quantities fixed. The solid curve is the baseline choice $w_{\rm coh} = 50\,\mu$m, and the shaded region is therefore a representative width-sensitivity band rather than an interquartile band from a full phenomenological transfer sweep. The figure should accordingly be read as a focused long-time thermal slice through the broader bridge law.

Additional exploratory checks suggest that varying the imported threshold ratio $\mathcal{I}_*/A$ across a broad representative range shifts the overall normalization less strongly than the coherence-cell bridge changes produced by LMT order, interrogation time, and the effective coherence/geometric ratios. The leading n^2 scaling and the clean regime separation between current $n \sim 10^2$ instruments and future $n \sim 10^3$ architectures therefore do not appear to rely sensitively on the specific benchmark choice $\mathcal{I}_*/A \approx 3.28$. The adoption of that value should accordingly be read primarily as a cross-program consistency choice rather than as the unique source of the laboratory forecast.

In the focused figure shown here, the plotted normalization varies only through the chosen width band in $w_{\rm coh}$, so the shaded region reflects the single composite factor $(w_{\rm coh}/\lambda_{\rm coh,0})^2$ under the thermal identification rather than a simultaneous sweep over several independent nuisance quantities.

For clarity, the shaded region in figure 1 is not a statistical confidence interval and does not represent the full phenomenological uncertainty of the bridge. It is a model-side width-sensitivity band at fixed T, Θ, $k_{\rm eff}$, χ, η, $\mathcal{I}_*/A$, and $\xi_{\rm th}$. Broader uncertainty associated with source/coherence transfer and residual order-unity geometric factors remains encoded in the bridge law itself, but is not simultaneously displayed in this focused illustration.

The benchmark analysis and the focused thermal width-sensitivity illustration support the same qualitative conclusion. The operational coherence-cell bridge does not predict a universal constant dephasing floor, but rather a residual geometric burden that grows as $Tn^2k_{\rm eff}^2$, with current $n \sim 10^2$ devices expected to remain significantly below the dephasing regime accessible to future $n \sim 10^3$ architectures.

For experimental purposes, the residual contrast-loss law may be fit at fixed interrogation time by

$$D_{\rm pred}(n) \;=\; a_0 + a_1 n^2, \tag{13}$$

where a_0 represents an approximately n-independent residual baseline offset over the scan range after standard technical-noise and environmental modeling, while a_1 is the coefficient of the NDK-motivated quadratic component.

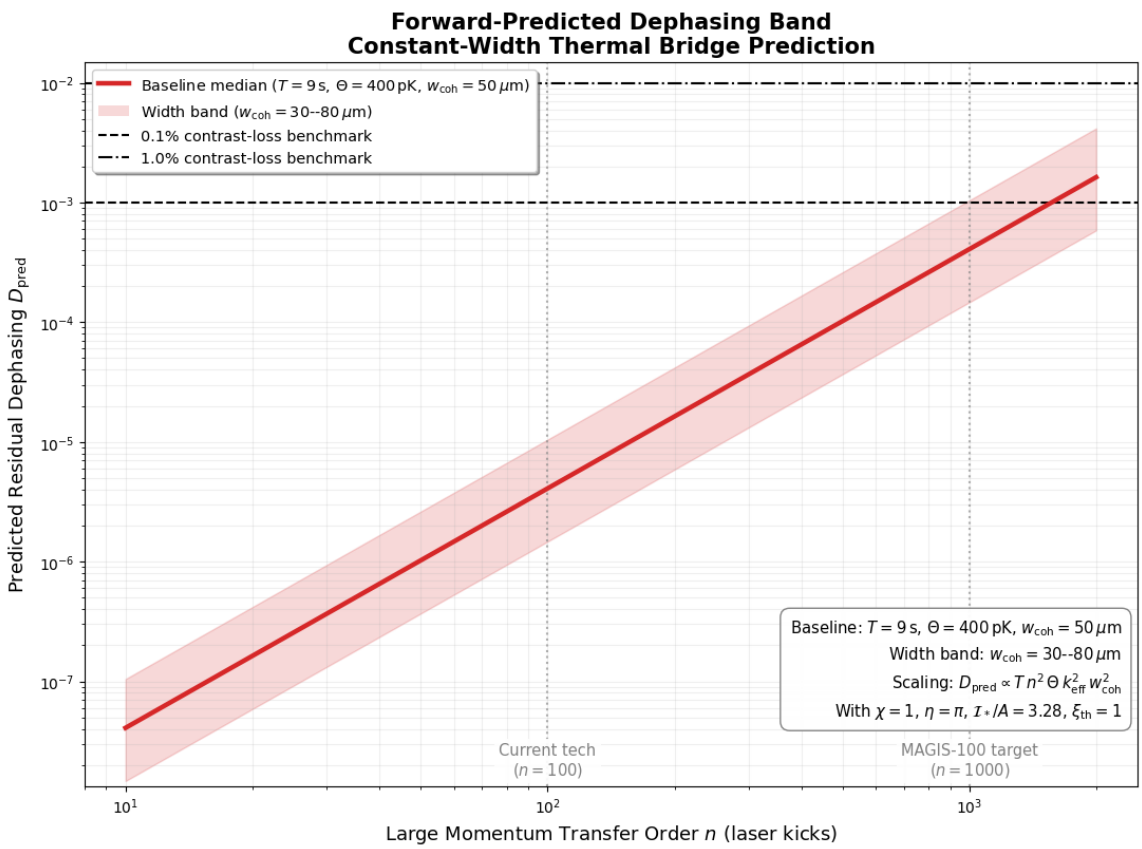

Figure 1: Forward-predicted dephasing band from the constant-width thermal bridge model. The solid curve shows the baseline prediction for $T = 9\,\mathrm{s}$, $\Theta = 400\,\mathrm{pK}$, and $w_{\mathrm{coh}} = 50\,\mu\mathrm{m}$, with $\chi = 1$, $\eta = \pi$, $\mathcal{I}_*/A = 3.28$, $\xi_{\mathrm{th}} = 1$, and $k_{\mathrm{eff}} = 10^7\,\mathrm{m}^{-1}$. The shaded region shows a representative width-sensitivity band obtained by varying w_{coh} over 30–80 μm while holding the other quantities fixed. The prediction preserves the characteristic n^2 scaling of the LMT transport law, with the overall normalization set by the chosen long-time thermal operating point. Vertical reference lines at $n = 100$ and $n = 1000$ indicate current-tech and MAGIS-like regimes, while horizontal lines at 10^{-3} and 10^{-2} mark sub-percent and percent-level contrast-loss benchmarks. The shaded band is intended as a model-side width-sensitivity band at fixed operating point rather than as a statistical experimental uncertainty interval or a full multi-parameter bridge sweep.

3.1 Temperature scaling under a thermal coherence identification

The operational coherence-cell bridge assumes that the pre-split coherence length is set by thermodynamic preparation. As a working identification, take the source coherence scale to be proportional to the thermal de Broglie scale,

$$\lambda_{\mathrm{coh},0}^2 = \xi_{\mathrm{th}}^2\,\lambda_{\mathrm{th}}^2, \qquad \lambda_{\mathrm{th}}^2 = \frac{2\pi\hbar^2}{m\,k_B\Theta}, \tag{14}$$

where Θ is the source temperature, m is the atomic mass, and ξ_{th} is an order-unity bridge factor allowing for the possibility that the operational pre-split coherence length is only proportional to the ideal thermal scale rather than exactly equal to it.

This thermal identification is no longer wholly external to the NDK reduction. In the reduced finite-temperature sector derived in [8], the Fisher coefficient naturally defines a thermal coherence scale with the standard thermal de Broglie form. The proportionality $\lambda_{\mathrm{coh},0} = \xi_{\mathrm{th}}\lambda_{\mathrm{th}}$ adopted here should therefore be read as a phenomenological bridge from that reduced-sector preparation scale to the operational pre-split source coherence, rather than as an unrelated ansatz.

Substituting equation (14) into the master phenomenological equation (9) gives

$$D_{\mathrm{pred}}(\Theta) = \frac{\chi}{2\eta\,\xi_{\mathrm{th}}^2}\,\mathcal{I}_*/A\,c\,\ell_{\mathrm{P}}\,T\,n^2 k_{\mathrm{eff}}^2\,w_{\mathrm{coh}}^2\,\frac{m\,k_B\Theta}{\hbar^2}. \tag{15}$$

In the fit-law language of Eq. (13), the quadratic dephasing coefficient therefore satisfies

$$a_1(T,\Theta) \propto T\,\Theta \tag{16}$$

at fixed geometry and fixed transverse coherence-support width.

This temperature scaling is specific to the NDK coherence-cell bridge: standard quantum mechanics by itself does not predict any residual geometric dephasing floor. The dependence on Θ arises because the present bridge ties the pre-split coherence cell to the thermal state of the source, so that a hotter source corresponds to a smaller coherence cell and therefore a larger effective transport burden at fixed LMT order.

The experimentally cleanest form is the ratio test. For two runs at source temperatures Θ_1 and Θ_2, with matched geometry, matched preparation protocol, and controlled transverse coherence-support width, one obtains

$$\frac{a_1(\Theta_2)}{a_1(\Theta_1)} = \frac{\Theta_2}{\Theta_1}. \tag{17}$$

Within the thermal-coherence identification above, this cancels the imported directional-strain benchmark together with the geometric and coherence-bridge normalizations, leaving a sharp scaling test of the bridge model. If the transverse coherence-support width is not held fixed across the temperature scan, the more general relation is

$$\frac{a_1(\Theta_2)}{a_1(\Theta_1)} = \frac{\Theta_2\,w_{\mathrm{coh}}^2(\Theta_2)}{\Theta_1\,w_{\mathrm{coh}}^2(\Theta_1)}, \tag{18}$$

which provides a direct diagnostic of whether w_{coh} has been successfully controlled.

3.2 The full experimental discriminator

Synthesizing these elements, the LMT-order scan remains the foundational experimental test. Technical noise may have many apparatus-specific dependencies, but the NDK residual forecast singles out a simple leading structure: after standard technical-noise and environmental modeling, the residual law isolates a leading quadratic dependence on n at fixed T, with the fitted coefficient a_1 scaling linearly with both interrogation time T and source temperature Θ.

More explicitly, the experimental discriminator is not the mere appearance of additional contrast loss, but the emergence—after standard technical-noise and environmental modeling—of a residual law whose leading dependence is quadratic in n at fixed T, and whose fitted quadratic coefficient scales linearly with T across runs with different interrogation times. Within the coherence-cell bridge developed here, the normalization of this residual law is then compared against the benchmark band shown in figure 1, with $\mathcal{I}_*/A$ held fixed at the cosmologically imported value rather than re-fit from laboratory data. If such a residual pattern appears, the experiment will not merely have seen

unexplained contrast loss; it will have isolated the specific n^2 geometry predicted by the closure-informed source-coherence transfer law.

Conversely, if future LMT interferometers operating in the relevant T, n, and coherence regime fail—after standard technical-noise and environmental modeling—to exhibit a residual contrast-loss law whose leading dependence is quadratic in n at fixed T, with the fitted quadratic coefficient scaling linearly with T, then the coherence-cell bridge proposed here is falsified. Similarly, if the quadratic coefficient fails to scale linearly with source temperature at fixed geometry and controlled w_{coh}, the thermal identification underlying the source-level coherence transfer law is independently falsified.

4 Conclusion

This Letter has presented a compact phenomenological bridge between the NDK transport framework and LMT atom interferometry. Recent work deriving emergent Schrödinger dynamics from the NDK master action has fixed the upstream matter-wave transport law and the quadratic phase-gradient burden as structural consequences of the reduced Fisher transport sector [8]. Recent action-scale closure work has further isolated the primitive-cell sector as the unique carrier of the absolute action normalization, removing the need to regard the Planck-anchored transport scale as a free reduced-sector ambiguity [9]. In the present Letter, the remaining major phenomenological step is therefore narrower than before: the coarse-graining from closure-informed substrate normalization to the experimentally active source-level coherence cell of a concrete interferometer. The bridge is accordingly built from a derived cell-level transport burden together with an operational pre-split coherence-cell bridge.

Two conclusions follow.

First, the operational coherence-cell bridge establishes a strict physical separation between the closure-informed primitive action scale of the substrate and the source-level operational coherence cell probed experimentally. This is what protects the n^2 signature and makes the proposal experimentally meaningful.

Second, the benchmark analysis and the focused thermal width-sensitivity illustration together indicate a more nuanced experimental picture than a simple large-loss forecast. Current $n \sim 10^2$ instruments remain parametrically below the relevant window, while long-time $n \sim 10^3$ architectures can move into the sub-percent sensitivity regime and, for broader representative coherence-support widths, approach the 10^{-3} contrast-loss benchmark. Because the leading signature is controlled primarily by the $T\,n^2 k_{\mathrm{eff}}^2$ structure of the coherence-cell bridge, the most important experimental opportunity is not necessarily an overwhelmingly large residual at nominal settings, but the appearance of a structured n^2 law under controlled sweeps of LMT order, interrogation time, and source preparation.

> *If the closure-informed operational coherence-cell bridge is the correct laboratory realization of NDK transport, then future $n \sim 10^3$ LMT interferometers should exhibit a residual contrast-loss component whose leading dependence is quadratic in LMT order, with long-time thermal operating points moving into the sub-percent sensitivity window and broader representative coherence-support widths approaching the 10^{-3} benchmark.*

That forecast is sharp enough to be falsifiable. If LMT atom interferometers operating in the relevant T, n, and coherence regime do not reveal a residual contrast-loss law whose leading dependence is quadratic in n at fixed T, with the quadratic coefficient scaling linearly with T, then

the coherence-cell bridge developed here is falsified as the proposed laboratory realization of NDK transport.

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